

How to use QR codes for BREATHE activities at sites

Purpose: This is a guide to help personnel at sites to use the QR codes for the BREATHE activities correctly.

QR codes (Figure 1) are the means of uniquely identifying clients in the BREATHE data. QR codes come in the form of stickers, and they are used on:

- The TB request forms or vouchers at the level of screening,
- Sputum mugs at the level of specimen collection/reception, and
- At the lab (microscopy, TB LAMP, Xpert) during data entry of test results.



Figure 1: QR code

- Using QR code can help to reduce the work of those doing the screening and those at the lab, as well as to increase the accuracy of linking screening to results for individual clients.
- This JA will demonstrate how these two types of QR codes can be used.

NOTE: With the QR code system, the QR code is the only thing that will link the client from the screening form to the sputum collection form and the lab result forms in the REDCap database. It is therefore essential that the QR code goes along with the same client from the OPD to the lab for clients that are eligible for a TB test.

- The BBREATHE QR codes are divided into two types: **QR codes for eligible client** and **QR codes for non-eligible clients**.
 - **Eligible QR codes:** It has 6 unique QR codes in a row printed for a unique client that is eligible for a TB test (Figure 2). Example: 1FOU-03121



Figure 2 : Ex. QR codes for eligible clients.

- **Non-Eligible QR codes:** non-eligible QR code is differentiated from an eligible QR code by an “x” at the end use for patient who do not need follow up. Each of these QR codes belongs to a single non-eligible client (figure 3). Example **1FOU-04220x**



Figure 3 : Ex. QR codes for non-eligible clients

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IMPORTANT: When working with QR codes, to avoid repeating the same QR code with a different client, the QR codes should be kept and used in chronological order, respecting the order as received. E.g. **1FOU-00001** for the 1st client, **1FOU-00002** for the 2nd client, **1FOU-00003** for the 2nd client referred for a test (see Figure 2 above).

I. FOR CLIENTS WHO ARE ELIGIBLE FOR TB TEST

A. Using the QR codes during screening (when a TB test is requested for the client screened)

IMPORTANT:

- When giving a TB test request form with a QR code or a sputum mug with a QR code to a person recommended to make a TB test, inform them that this is what they will use to get their free TB test, so they understand that they do not have to pay for the test.
- Remind the person that sample submission for testing does not take much time.
- When sticking the QR codes, either on the request forms, vouchers or sputum mug, **ensure that it is not twisted but neatly and straightly done.**

1. If the client is eligible for a TB test after screening, fill a TB test request form (or voucher) then peel off the first QR code in the row in the sequence of QR codes and stick this onto the request form or voucher. Scan the QR code in the WAVE 11 Project Primary Health Care project in the REDCap app in the required field. **IMPORTANT:** Ensure that the QR code that appears on the phone after scanning matches the QR code scanned (QR code on the request form, for the client in front of you).

In the event where the code which appears in the phone does not match the QR code scanned, proceed to the next code and cross out the line of code that has been skipped. Remember to take a snapshot of the mis-matching QR codes and report this to your zone coordinator.

2. Take a sputum mug, peel off 2 of the same QR codes on the same row (make sure it is the same QR code that was used on the request form/voucher) stick 1 to the lid of the sputum mug and the other corresponding sticker on the side of the sputum mug (if 2 samples are requested, also stick 2 QR codes on the 2nd sputum mug). See Fig. 4.



Figure 4: Ex. of labeled sputum mug with QR code and Xpert pooled testing category

3. If the BREATHE REDCap screening form indicates that the person has a high, medium, or low risk for TB, then use a marker to write this information on the sputum mug. (This will be used by the Xpert testing lab to decide the pool size). See Fig. 4.
4. Give the participants sputum production instructions either using the spontaneous sputum production video (s) or demonstrate to clients on how to produce a good sputum. For clients who may face difficulties producing a sample, you can use the Lung Flute ECO to help them produce a sample.

NOTE: Sputum samples should be produced in an open area, away from the general population or in a designated area. There may be a biosafety and infection risk if people take the mug and produce sputum in the OPD or anywhere in the hospital, rather than in the dedicated space to produce sputum. We need to be careful about giving appropriate instructions.

5. Once done, handover the sputum mug to the client, indicate the location to produce the sputum and instruct them to return with it immediately done.

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6. Repeat steps 1 to 5 for the next clients referred for a TB test.

B. Using the QR codes to send samples for TB testing

7. For samples that are to be sent for testing (microscopy (DTC) and/or Xpert testing site), gather the samples collected for the day and arrange the samples in ascending QR code number found on the sputum mugs.
8. Fill the “Sputum Sending List” with the necessary information, i.e., Specimen ID (QR Code ID), sex, age, Date of collection, initial results (microscopy and/or TB LAMP-DTCs) corresponding to the samples collected for the day for sending, as well as the risk for TB (high, medium, low), if indicated during screening,
9. Package the samples respecting the triple packaging of sputum samples for transportation and place in the specimen flask, alongside the corresponding request form/voucher for each specimen + QR code, which will be collected by the transporter.

C. Using the QR codes at the lab for lab result data entry

10. For those carrying out the TB test (local lab and/or Xpert testing site) when entering the participant’s information into the lab register, indicate the client’s QR code beside the entry (or as follow instructions as indicated in the job aid for Xpert pooled testing), at a position where it can be easily seen.
11. To enter participant lab results into the REDCap app, get the client’s request form. First enter the clients’ results, from the testing register, onto the request form, then proceed to entering the results in the REDCap app.
12. Access the required form (Local Lab Result Form-Microscopy/Urine LF-LAM, TB LAMP or Ultra Results) in the Wave 11 LAB RESULT FORMSproject in the REDCap app, then scan the QR code in the app, and proceed with entering the results of the client.

II. FOR CLIENTS WHO ARE NOT ELIGIBLE FOR A TB TEST

13. Identify the QR code used for non-eligible clients (figure 3). This should be used for all clients that are screened and are not referred for a TB test.
14. Peel off first QR code neatly and stick on a clean background (example white cardboard paper).
15. Scan the QR code in the scanned QR code field provided.
16. Use a Red marker to strike off the used QR code scanned as seen in figure 5
17. Repeat step 13 – 16 for the next non- eligible client screened with the WAVE 11 Project Primary Health Careproject.
18. Clear off the cardboard at the end of the day and discard all the used non-eligible QR codes.



Figure 5: How to place QR codes for non- eligible clients before scanning.